GMAT Quantitative Reasoning

Percents, Ratios & Proportions

Hard / Very Hard Problem Modeling

Challenging exercises focused on modeling complex problems involving percentages, ratios, and proportions. These problems require careful analysis, multi-step reasoning, and awareness of common traps. Work through the questions first, then check the detailed solutions.

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

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1. Part I: Exercises

Work through each problem carefully. Focus on setting up the problem correctly before attempting to solve it. These problems are designed to challenge your problem-modeling skills at a high level.

Difficulty Level: Hard to Very Hard

These problems require:

- Careful handling of successive percent changes
- Deep understanding of ratio relationships and scaling
- Multi-step reasoning with proportions
- Awareness of common GMAT traps in percentage problems

1.1. Section A: Advanced Percentage Problems

1.1.1. Exercise A1: Successive Percent Changes

A store increases the price of an item by 20%, then decreases the new price by 20%.

Questions:

1. What is the net percent change from the original price?
2. If the original price was $100, what is the final price?
3. What single percent decrease would be equivalent to "increase by p%, then decrease by p %"? Express in terms of p.
4. A price is increased by 25% and then decreased by x% to return to the original price. Find x.

1.1.2. Exercise A2: Compound Percentage Growth

A population grows by 10% in the first year and by 20% in the second year.

Questions:

1. What is the total percent increase over the two years?
2. If the population after two years is 26,400, what was the original population?
3. If the population had grown by the same rate each year to achieve the same final population, what would that constant annual rate be?
4. After a third year, the population is 34,320. What was the percent change in the third year?

1.1.3. Exercise A3: Percent of Percent

In a survey, 60% of respondents are female. Of the female respondents, 40% are under 30 years old. Of the male respondents, 25% are under 30 years old.

Questions:

1. What percent of all respondents are females under 30?
2. What percent of all respondents are under 30?
3. What percent of respondents under 30 are female?
4. If there are 500 total respondents, how many males are 30 or older?

1.1.4. Exercise A4: Percentage with Constraints

A store offers a 30% discount on all items. Additionally, members receive an extra 10% off the discounted price. Sales tax of 8% is applied after all discounts.

Questions:

1. If an item originally costs $200, what is the final price for a member (including tax)?
2. What single discount percentage would be equivalent to the combined 30% and 10% member discount (before tax)?
3. A member pays $113.40 including tax. What was the original price?
4. At what original price would a member pay exactly $100 after all discounts and tax?

1.1.5. Exercise A5: Percent Distribution

In a company, employees are distributed across three departments: Sales, Engineering, and Marketing. Sales has 40% of employees, Engineering has 35%, and Marketing has the rest.

Questions:

1. What percent of employees are in Marketing?
2. If 30% of Sales employees and 20% of Engineering employees work remotely, what percent of the company works remotely if no Marketing employees work remotely?
3. If 10% of the total workforce is transferred from Sales to Engineering, what are the new percentages for each department?
4. The company hires new employees, increasing the Engineering department by 40% while keeping Sales and Marketing unchanged. What percent of the new total workforce is in Engineering?

1.2. Section B: Ratio Problems

1.2.1. Exercise B1: Combining Ratios

The ratio of boys to girls in Class A is 3:2. The ratio of boys to girls in Class B is 4:5.

Questions:

1. If Class A has 30 students and Class B has 45 students, how many boys are there in total?
2. What is the combined ratio of boys to girls across both classes?
3. If the total number of students is 120 and the combined ratio of boys to girls is 11:13, how many students are in each class?
4. What ratio of Class A students to Class B students would make the overall boy-to-girl ratio exactly 1:1?

1.2.2. Exercise B2: Ratio Changes

The ratio of red balls to blue balls in a bag is 5:3.

Questions:

1. If there are 40 balls in total, how many are red?
2. If 4 red balls are removed and 4 blue balls are added, what is the new ratio?
3. How many blue balls must be added (without removing any red) to make the ratio 1:1?
4. Starting from the original 40 balls, if n red balls are removed, the ratio becomes 1:1. Find n.

1.2.3. Exercise B3: Part-to-Whole Ratios

A fruit basket contains apples, oranges, and bananas in the ratio 2:3:5.

Questions:

1. What fraction of the basket is oranges?
2. If there are 24 oranges, how many pieces of fruit are in the basket?
3. The ratio of apples to the total fruit is what percent?
4. If 10 more apples are added, the new ratio of apples to oranges is 1:1. How many of each fruit were there originally?

1.2.4. Exercise B4: Ratio Word Problems

In a school, the ratio of teachers to students is 1:15. The ratio of male teachers to female teachers is 2:3.

Questions:

1. If there are 450 students, how many teachers are there?
2. How many female teachers are there?
3. If 5 new male teachers are hired, what is the new ratio of male to female teachers?
4. If the school wants the ratio of teachers to students to be 1:12, how many new teachers must be hired (assuming the number of students stays the same)?

1.3. Section C: Proportion Problems

1.3.1. Exercise C1: Direct and Inverse Proportion

If 8 workers can complete a project in 15 days, assume work rate is constant.

Questions:

1. How many days would 12 workers take to complete the same project?
2. How many workers are needed to complete the project in 10 days?
3. If after 5 days of work by 8 workers, 4 more workers join, how many total days does the project take?
4. If the project is twice as large, how many workers are needed to complete it in 20 days?

1.3.2. Exercise C2: Scale Factors

A map has a scale of 1:50,000 (1 cm on the map represents 50,000 cm in reality).

Questions:

1. If two cities are 8 cm apart on the map, what is the actual distance in kilometers?
2. A park has an area of 2 cm^2 on the map. What is the actual area in square kilometers?
3. A rectangular region on the map measures 3 cm by 4 cm. What is the actual perimeter in kilometers?
4. If the scale were changed to 1:25,000, what would be the new map distance between the two cities that are actually 4 km apart?

1.3.3. Exercise C3: Proportional Reasoning

The cost of a carpet is proportional to its area. A 4m by 5m carpet costs $600.

Questions:

1. What is the cost per square meter?
2. How much would a 3m by 8m carpet cost?
3. If the length and width of a carpet are both increased by 50%, by what percent does the cost increase?
4. A customer has a budget of $1,000. What is the maximum area of carpet they can buy?

1.3.4. Exercise C4: Three-Variable Proportions

The time T to fill a pool is directly proportional to the volume V and inversely proportional to the flow rate R.

Questions:

1. Write an equation relating T, V, and R with proportionality constant k.
2. If a 1000-liter pool fills in 2 hours at a rate of 500 liters/hour, find k.
3. How long would it take to fill a 1500-liter pool at a rate of 600 liters/hour?
4. If the volume is doubled and the flow rate is tripled, by what factor does the time change?

1.4. Section D: Complex Percent and Ratio Applications

1.4.1. Exercise D1: Profit and Loss

A merchant buys an item for $80 and marks it up by 50% to set the selling price. He then offers a 20% discount on the marked price.

Questions:

1. What is the marked price?
2. What is the selling price after the discount?
3. What is the merchant's profit or loss, and what percent is this of the cost price?
4. What discount percentage on the marked price would result in a 10% profit on the cost price?

1.4.2. Exercise D2: Mixture Ratios

Solution A is 20% acid and Solution B is 60% acid.

Questions:

1. In what ratio should A and B be mixed to get a 35% acid solution?
2. If 200 mL of A is mixed with 300 mL of B, what is the concentration of the resulting mixture?
3. How much of Solution B must be added to 400 mL of Solution A to create a 50% acid solution?
4. If you have 500 mL of a 35% acid solution and want to increase its concentration to 45% by adding Solution B, how much of B is needed?

1.4.3. Exercise D3: Interest Rate Problems

An investment of $10,000 earns simple interest at an annual rate of 5%.

Questions:

1. How much interest is earned after 3 years?
2. After how many years will the investment double (simple interest)?
3. If the same $10,000 earns compound interest at 5% annually, how much is the investment worth after 3 years?
4. What simple interest rate would yield the same amount after 3 years as 5% compound interest?

1.4.4. Exercise D4: Population and Percentage Change

City X has a population of 100,000 and City Y has a population of 80,000. City X's population decreases by 5% per year, while City Y's population increases by 10% per year.

Questions:

1. What are the populations after 1 year?
2. After how many complete years will City Y's population first exceed City X's?
3. If both growth rates continue, what is the ratio of City Y's population to City X's after 2 years?
4. At what constant annual rate would City X need to grow for both cities to have equal populations after 2 years?

1.5. Section E: Unit Conversions and Dimensional Analysis

1.5.1. Exercise E1: Speed and Distance Conversions

A car travels at 72 km/h.

Questions:

1. Convert this speed to meters per second.
2. How far does the car travel in 2.5 hours (in kilometers)?
3. If the car travels for 45 minutes, how many meters does it cover?
4. At what speed in km/h would the car need to travel to cover 100 km in 48 minutes?

1.5.2. Exercise E2: Area and Volume Conversions

A rectangular room measures 4.5 m by 6 m with a ceiling height of 3 m.

Questions:

1. What is the floor area in square centimeters?
2. What is the volume of the room in cubic meters?
3. If paint covers 10 m^2 per liter, how many liters are needed to paint all four walls (no windows or doors)?
4. If carpet costs $25 per square foot, how much would it cost to carpet the floor? (1 m $\approx$ 3.28 ft)

1.5.3. Exercise E3: Rate Conversions

A factory produces 240 units per hour.

Questions:

1. How many units are produced per minute?
2. How many units are produced in an 8-hour shift?
3. If the factory operates 5 days a week for 8 hours per day, how many units are produced per week?
4. At what rate (units per hour) would the factory need to operate to produce 10,000 units in a 5-day work week (8 hours per day)?

1.5.4. Exercise E4: Currency and Exchange Rates

The exchange rate is 1 USD = 0.85 EUR and 1 EUR = 130 JPY.

Questions:

1. How many EUR is 500 USD worth?
2. How many JPY is 500 USD worth?
3. If an item costs 15,000 JPY, what is its price in USD?
4. If the USD/EUR rate changes to 1 USD = 0.90 EUR (while EUR/JPY stays the same), by what percent does the USD/JPY rate change?

2. Part II: Solutions

2.1. Section A Solutions: Advanced Percentage Problems

2.1.1. Solution A1: Successive Percent Changes

Question 1: Net percent change:

Let original price $= 100$ After 20% increase: $100 \times 1.20 = 120$ After 20% decrease: $120 \times 0.80 = 96$ Net change $= 96 - 100 = -4$, which is -4%

Answer: The net percent change is -4% (a decrease).

Question 2: Final price if original is $100:

Final price $= \$96$ (from calculation above)

Answer: The final price is $96.

Question 3: General formula for increase by p%, then decrease by p%:

Multiplier $= \left(1 + \frac{p}{100}\right)\left(1 - \frac{p}{100}\right) = 1 - \left(\frac{p}{100}\right)^2$

Net change $= -\frac{p^2}{10000} = -\left(\frac{p^2}{100}\right)\%$

Answer: The equivalent single percent decrease is $\left(\frac{p^2}{100}\right)\%$ or $\frac{p^2}{10000}$ as a decimal.

Question 4: Find x if 25% increase followed by x% decrease returns to original:

$1.25 \times \left(1 - \frac{x}{100}\right) = 1$ $1 - \frac{x}{100} = \frac{1}{1.25} = 0.80$ $\frac{x}{100} = 0.20$ $x = 20$

Answer: $x = 20\%$

2.1.2. Solution A2: Compound Percentage Growth

Question 1: Total percent increase over two years:

Multiplier $= 1.10 \times 1.20 = 1.32$ Total increase $= 32\%$

Answer: The total percent increase is 32%.

Question 2: Original population:

Let original $= P$ $P \times 1.32 = 26400$ $P = \frac{26400}{1.32} = 20000$

Answer: The original population was 20,000.

Question 3: Constant annual rate:

Let rate $= r$ $(1 + r)^2 = 1.32$ $1 + r = \sqrt{1.32} \approx 1.1489$ $r \approx 0.1489 = 14.89\%$

Answer: The constant annual rate would be approximately 14.89%.

Question 4: Third year percent change:

Year 2 population $= 26400$ Year 3 population $= 34320$ Change $= \frac{34320}{26400} = 1.30$ Percent change $= 30\%$

Answer: The population increased by 30% in the third year.

2.1.3. Solution A3: Percent of Percent

Question 1: Females under 30:

$60\% \times 40\% = 0.60 \times 0.40 = 0.24 = 24\%$

Answer: 24% of all respondents are females under 30.

Question 2: Total under 30:

Females under 30: 24% Males: 40% of total Males under 30: $40\% \times 25\% = 10\%$ Total under 30: $24\% + 10\% = 34\%$

Answer: 34% of all respondents are under 30.

Question 3: Percent of under-30 who are female:

$24\frac{\%}{34}\% = \frac{24}{34} = \frac{12}{17} \approx 70.6\%$

Answer: Approximately 70.6% of respondents under 30 are female.

Question 4: Males 30 or older:

Total males $= 40\%$ of $500 = 200$ Males under 30 $= 10\%$ of $500 = 50$ Males 30 or older $= 200 - 50 = 150$

Answer: 150 males are 30 or older.

2.1.4. Solution A4: Percentage with Constraints

Question 1: Final price for member:

Original: $200 After 30% discount: $200 \times 0.70 = \$140$ After additional 10% member discount: $140 \times 0.90 = \$126$ After 8% tax: $126 \times 1.08 = \$136.08$

Answer: The final price is $136.08.

Question 2: Combined discount:

Combined multiplier $= 0.70 \times 0.90 = 0.63$ Equivalent single discount $= 1 - 0.63 = 0.37 = 37\%$

Answer: The equivalent single discount is 37%.

Question 3: Original price if member pays $113.40:

Let original $= P$ $P \times 0.63 \times 1.08 = 113.40$ $P \times 0.6804 = 113.40$ $P = \frac{113.40}{0.6804} = \$166.67$

Answer: The original price was $166.67.

Question 4: Original price to pay exactly $100:

$P \times 0.6804 = 100$ $P = \frac{100}{0.6804} \approx \$147.00$

Answer: The original price should be approximately $147.00.

2.2. Section B Solutions: Ratio Problems

2.2.1. Solution B1: Combining Ratios

Question 1: Total boys:

Class A (30 students, ratio 3:2): Boys $= 30 \times \frac{3}{5} = 18$ Class B (45 students, ratio 4:5): Boys $= 45 \times \frac{4}{9} = 20$ Total boys $= 18 + 20 = 38$

Answer: There are 38 boys in total.

Question 2: Combined ratio:

Total boys $= 38$ Total girls $= (30 - 18) + (45 - 20) = 12 + 25 = 37$ Combined ratio $= 38 : 37$

Answer: The combined ratio is 38:37.

Question 3: Students in each class if combined ratio is 11:13:

Boys $= 120 \times \frac{11}{24} = 55$ Girls $= 120 \times \frac{13}{24} = 65$

Let Class A have a students, Class B have $120 - a$ students. Boys from A: $3\frac{a}{5}$ Boys from B: $4\frac{120-a}{9}$ $3\frac{a}{5} + 4\frac{120-a}{9} = 55$

Multiplying by 45: $27a + 20(120 - a) = 2475$ $27a + 2400 - 20a = 2475$ $7a = 75$ $a = \frac{75}{7} \approx 10.7$

This doesn't give integer values. Let me recheck…

Actually, the constraint may not yield integer class sizes. Class A ≈ 10.7, Class B ≈ 109.3.

Answer: The constraints do not yield integer class sizes with the given combined ratio.

Question 4: Ratio for overall 1:1:

Let Class A have $5k$ students (3k boys, 2k girls) Let Class B have $9m$ students (4m boys, 5m girls) For 1:1 overall: $3k + 4m = 2k + 5m$ $k = m$

Ratio of A to B: $5k : 9m = 5 : 9$

Answer: Class A to Class B ratio should be 5:9.

2.2.2. Solution B2: Ratio Changes

Question 1: Red balls out of 40:

Ratio 5:3 means 8 parts total. Red $= 40 \times \frac{5}{8} = 25$

Answer: There are 25 red balls.

Question 2: New ratio after removing 4 red, adding 4 blue:

New red $= 25 - 4 = 21$ New blue $= 15 + 4 = 19$ New ratio $= 21 : 19$

Answer: The new ratio is 21:19.

Question 3: Blue balls to add for 1:1 ratio:

Original: 25 red, 15 blue For 1:1: need blue $= 25$ Add $= 25 - 15 = 10$ blue balls

Answer: 10 blue balls must be added.

Question 4: Red balls to remove for 1:1:

After removing n red: $(25 - n)$ red, 15 blue $(25 - n) = 15$ $n = 10$

Answer: $n = 10$ red balls must be removed.

2.2.3. Solution B3: Part-to-Whole Ratios

Question 1: Fraction that is oranges:

Total parts $= 2 + 3 + 5 = 10$ Oranges $= \frac{3}{10}$

Answer: Oranges are $\frac{3}{10}$ of the basket.

Question 2: Total fruit if 24 oranges:

Oranges are $\frac{3}{10}$ of total. $24 = \left(\frac{3}{10}\right) \times \text{Total}$ Total $= 24 \times \frac{10}{3} = 80$

Answer: There are 80 pieces of fruit.

Question 3: Apples as percent of total:

Apples $= \frac{2}{10} = \frac{1}{5} = 20\%$

Answer: Apples are 20% of the total.

Question 4: Original quantities if adding 10 apples makes apples:oranges = 1:1:

Let original have $2k$ apples, $3k$ oranges, $5k$ bananas. After adding 10: $2k + 10$ apples $2k + 10 = 3k$ (for 1:1 with oranges) $k = 10$

Original: Apples $= 20$, Oranges $= 30$, Bananas $= 50$

Answer: Originally: 20 apples, 30 oranges, 50 bananas.

2.3. Section C Solutions: Proportion Problems

2.3.1. Solution C1: Direct and Inverse Proportion

Question 1: Days for 12 workers:

Work $= 8 \times 15 = 120$ worker-days Days for 12 workers $= \frac{120}{12} = 10$

Answer: 12 workers would take 10 days.

Question 2: Workers for 10 days:

Workers $= \frac{120}{10} = 12$

Answer: 12 workers are needed.

Question 3: Total days with workers joining:

Work done in first 5 days: $8 \times 5 = 40$ worker-days Remaining work: $120 - 40 = 80$ worker-days Workers after day 5: $8 + 4 = 12$ Days for remaining: $\frac{80}{12} = \frac{20}{3} \approx 6.67$ days Total: $5 + 6.67 = 11.67$ days

Answer: The project takes approximately 11.67 days (or $\frac{35}{3}$ days exactly).

Question 4: Workers for double project in 20 days:

Work $= 2 \times 120 = 240$ worker-days Workers $= \frac{240}{20} = 12$

Answer: 12 workers are needed.

2.3.2. Solution C2: Scale Factors

Question 1: Actual distance:

$8 \text{ cm} \times 50000 = 400000 \text{ cm} = 4000\text{m} = 4 \text{ km}$

Answer: The actual distance is 4 km.

Question 2: Actual area:

Scale for area $= 50000^2 = 2.5 \times 10^9$ Actual area $= 2\text{cm}^2 \times 2.5 \times 10^9 = 5 \times 10^9\text{cm}^2 = 5 \times 10^5\text{m}^2 = 0.5\text{km}^2$

Answer: The actual area is 0.5 km² (or 500,000 m²).

Question 3: Actual perimeter:

Map perimeter $= 2(3+4) = 14$ cm Actual perimeter $= 14 \times 50000 = 700000$ cm $= 7$ km

Answer: The actual perimeter is 7 km.

Question 4: New map distance at 1:25,000 scale:

Actual distance $= 4$ km $= 400000$ cm New map distance $= \frac{400000}{25000} = 16$ cm

Answer: The new map distance would be 16 cm.

2.3.3. Solution C3: Proportional Reasoning

Question 1: Cost per square meter:

Area $= 4 \times 5 = 20$ m² Cost per m² $= \frac{600}{20} = \$30$

Answer: The cost is $30 per square meter.

Question 2: Cost of 3m by 8m carpet:

Area $= 24$ m² Cost $= 24 \times 30 = \$720$

Answer: The carpet would cost $720.

Question 3: Percent increase if dimensions increase by 50%:

New dimensions: $1.5L \times 1.5W$ New area $= 2.25 \times$ original area Cost increases by 125%

Answer: The cost increases by 125%.

Question 4: Maximum area for $1,000:

Area $= \frac{1000}{30} = 33.33$ m²

Answer: Maximum area is 33.33 m² (or $\frac{100}{3}$ m²).

2.4. Section D Solutions: Complex Applications

2.4.1. Solution D1: Profit and Loss

Question 1: Marked price:

Cost $= \$80$ Markup $= 50\%$ Marked price $= 80 \times 1.50 = \$120$

Answer: The marked price is $120.

Question 2: Selling price after discount:

$120 \times 0.80 = \$96$

Answer: The selling price is $96.

Question 3: Profit percent:

Profit $= 96 - 80 = \$16$ Percent $= \frac{16}{80} = 20\%$

Answer: The merchant makes a 20% profit.

Question 4: Discount for 10% profit:

Target selling price $= 80 \times 1.10 = \$88$ Discount from marked price $= \frac{120-88}{120} = \frac{32}{120} = 26.67\%$

Answer: A discount of approximately 26.67% would yield 10% profit.

2.4.2. Solution D2: Mixture Ratios

Question 1: Ratio for 35% solution:

Let ratio be $A : B = x : y$ $\frac{0.20x+0.60y}{x+y} = 0.35$ $0.20x + 0.60y = 0.35x + 0.35y$ $0.25y = 0.15x$ $\frac{y}{x} = \frac{0.15}{0.25} = \frac{3}{5}$

Ratio $A : B = 5 : 3$

Answer: Mix A and B in ratio 5:3.

Question 2: Concentration of 200 mL A + 300 mL B:

Acid $= 200(0.20) + 300(0.60) = 40 + 180 = 220$ mL Concentration $= \frac{220}{500} = 44\%$

Answer: The resulting concentration is 44%.

Question 3: Amount of B to add to 400 mL of A for 50%:

Let x mL of B be added. $\frac{400(0.20)+x(0.60)}{400+x} = 0.50$ $80 + 0.60x = 200 + 0.50x$ $0.10x = 120$ $x = 1200$ mL

Answer: 1200 mL of Solution B must be added.

Question 4: B needed to increase 500 mL of 35% to 45%:

$\frac{500(0.35)+x(0.60)}{500+x} = 0.45$ $175 + 0.60x = 225 + 0.45x$ $0.15x = 50$ $x = 333.33$ mL

Answer: Approximately 333.33 mL of Solution B is needed.

2.4.3. Solution D3: Interest Rate Problems

Question 1: Simple interest after 3 years:

$I = P \times r \times t = 10000 \times 0.05 \times 3 = \$1500$

Answer: $1,500 in interest is earned.

Question 2: Years to double (simple):

$20000 = 10000 + 10000 \times 0.05 \times t$ $10000 = 500t$ $t = 20$ years

Answer: The investment doubles after 20 years.

Question 3: Compound interest after 3 years:

$A = 10000 \times (1.05)^3 = 10000 \times 1.157625 = \$11576.25$

Answer: The investment is worth $11,576.25.

Question 4: Simple rate equivalent to compound:

For simple interest to yield $11,576.25: $11576.25 = 10000(1 + r \times 3)$ $1 + 3r = 1.157625$ $r = 0.052542 = 5.25\%$

Answer: A simple interest rate of approximately 5.25% would be equivalent.

2.4.4. Solution D4: Population and Percentage Change

Question 1: Populations after 1 year:

City X: $100000 \times 0.95 = 95000$ City Y: $80000 \times 1.10 = 88000$

Answer: City X: 95,000; City Y: 88,000.

Question 2: Years until Y exceeds X:

After n years: $X = 100000 \times (0.95)^n$, $Y = 80000 \times (1.10)^n$

$Y > X$ when $80000 \times (1.10)^n > 100000 \times (0.95)^n$ $\left(\frac{1.10}{0.95}\right)^n > \frac{100000}{80000} = 1.25$ $(1.1579)^n > 1.25$

$n = 1$: $1.1579 < 1.25$ $n = 2$: $1.3407 > 1.25$ ✓

Answer: After 2 complete years, City Y's population first exceeds City X's.

Question 3: Ratio after 2 years:

X after 2 years: $100000 \times (0.95)^2 = 90250$ Y after 2 years: $80000 \times (1.10)^2 = 96800$ Ratio Y:X $= \frac{96800}{90250} \approx 1.073$

Answer: The ratio is approximately 1.073:1 or about 968:903.

Question 4: Growth rate for X to equal Y after 2 years:

$100000(1+r)^2 = 80000(1.10)^2 = 96800$ $(1+r)^2 = 0.968$ $1 + r = 0.9839$ $r = -0.0161 = -1.61\%$

X would still need to decrease (or grow at -1.61%).

Answer: City X would need to shrink by approximately 1.61% per year.

2.5. Section E Solutions: Unit Conversions

2.5.1. Solution E1: Speed and Distance Conversions

Question 1: Convert 72 km/h to m/s:

$72\ \mathrm{km/h} = 72 \times \frac{1000}{3600} = \frac{72}{3.6} = 20\ \mathrm{m/s}$

Answer: 72 km/h $= 20$ m/s.

Question 2: Distance in 2.5 hours:

$72 \times 2.5 = 180$ km

Answer: The car travels 180 km.

Question 3: Distance in 45 minutes:

45 min $= 0.75$ hours Distance $= 72 \times 0.75 = 54$ km $= 54000$ m

Answer: The car covers 54,000 meters.

Question 4: Speed to cover 100 km in 48 minutes:

48 min $= 0.8$ hours Speed $= \frac{100}{0.8} = 125$ km/h

Answer: The car would need to travel at 125 km/h.

2.5.2. Solution E2: Area and Volume Conversions

Question 1: Floor area in cm²:

$4.5\text{m} \times 6\text{m} = 27\text{m}^2 = 27 \times 10000\text{cm}^2 = 270000\text{cm}^2$

Answer: The floor area is 270,000 cm².

Question 2: Volume in m³:

$4.5 \times 6 \times 3 = 81\text{m}^3$

Answer: The volume is 81 m³.

Question 3: Paint for walls:

Wall areas: $2(4.5 \times 3) + 2(6 \times 3) = 27 + 36 = 63\text{m}^2$ Paint needed $= \frac{63}{10} = 6.3$ liters

Answer: 6.3 liters of paint are needed.

Question 4: Carpet cost in $/sq ft:

Floor area $= 27\text{m}^2$ In feet: $4.5 \times 3.28 = 14.76$ ft, $6 \times 3.28 = 19.68$ ft Area $= 14.76 \times 19.68 = 290.5\text{ft}^2$ Cost $= 290.5 \times 25 = \$7262.50$

Answer: The carpet would cost approximately $7,262.50.